Neck Anatomy: Contents & Clinical Correlations (Set 2)

1. A 24-year-old patient presents with a deep infection in the submandibular triangle. Surgical drainage must be performed carefully because running immediately across the superficial face of the submandibular gland is which motor nerve branch supplying the lower lip?
a. Temporal branch of CN VII
b. Zygomatic branch of CN VII
c. Buccal branch of CN VII
d. Marginal mandibular branch of the facial nerve
e. Cervical branch of CN VII
2. During extraoral surgical removal of the submandibular salivary gland within the digastric triangle, which major, highly tortuous blood vessel that grooves the posterior aspect of the gland must be ligated?
a. Maxillary artery
b. Lingual artery
c. Facial artery
d. Superior thyroid artery
e. Internal carotid artery
3. If a surgeon operates deeply near the floor of the submandibular triangle (medial to the hyoglossus muscle), they risk severing a major artery that provides the primary blood supply to the:
a. Upper teeth
b. Tongue (the lingual artery)
c. Parotid gland
d. External ear canal
e. Hard palate mucosa
4. A patient presents with a localized swelling in the root of the neck. Radiographic imaging confirms a mass compressing the contents of the supraclavicular triangle against the first rib, resulting in a loss of pulse in the upper limb. This artery is the:
a. Common carotid artery
b. Subclavian artery (third part)
c. Internal thoracic artery
d. Axillary artery
e. Vertebral artery
5. Penetrating trauma to the posterior triangle of the neck can slice through the investing layer of deep cervical fascia and sever the spinal accessory nerve (CN XI). This injury results in a complete paralysis of which muscle, causing drooping of the shoulder?
a. Sternocleidomastoid only
b. Trapezius muscle
c. Levator scapulae
d. Splenius capitis
e. Rhomboid major
6. A clinician performs a surgical block at the nerve point of the neck (Erb's point) along the mid-posterior border of the sternocleidomastoid muscle. This safely anesthetizes cutaneous branches derived from which neural network?

a. Cervical plexus (C2, C3, C4 segments)
b. Brachial plexus trunks
c. Cranial nerve terminal loops
d. Sympathetic chain ganglion roots
e. Intercostal rami

7. A child presents with a painless, fluctuating fluid-filled midline swelling in the anterior neck that moves upward when the child swallows or protrudes their tongue. This clinical presentation is characteristic of a:

a. Branchial cyst
b. Carotid body tumor
c. Thyroglossal duct cyst
d. Parotid mucocele
e. Dermoid tumor

8. A 45-year-old patient presents with a slow-growing, painless swelling at the anterior border of the sternocleidomastoid muscle near the angle of the jaw. Pathological evaluation confirms a developmental remnant of the second pharyngeal cleft, which is a:

a. Thyroglossal cyst
b. Dermoid cyst
c. Ranula
d. Branchial cyst (lateral cervical cyst)
e. Lingual thyroid node

9. A patient presents with a highly vascular mass located at the bifurcation of the common carotid artery inside the carotid triangle. Biopsy reveals a paraganglioma arising from chemoreceptor cells, formally diagnosed as a:

a. Carotid sinus aneurysm
b. Carotid body tumor (glomus tumor)
c. Internal jugular thrombosis
d. Thyroid adenoma
e. Lymphoma node

10. Digital pressure applied over the carotid sinus baroreceptors within the carotid triangle can cause a profound physiological drop in heart rate and blood pressure. This mechanical phenomenon is clinically termed:

a. Carotid sinus syncope (or reflex bradycardia)
b. Vasomotor shock
c. Hyperventilatory collapse
d. Epistaxis arrest
e. Trismus relaxation

11. During a central venous line placement, a clinician introduces a catheter into the internal jugular vein within the carotid triangle. The needle must be directed carefully to avoid piercing which major vessel situated immediately medial to the vein?

a. Subclavian vein
b. Common carotid artery (or internal carotid artery)
c. External jugular vein
d. Vertebral artery
e. Superficial temporal artery

12. An emergency cricothyroidotomy is performed to establish an airway when the glottis is completely obstructed. The incision is safely made through the skin and the cricothyroid membrane, which is located between the:
a. Hyoid bone and thyroid cartilage
b. Thyroid cartilage and cricoid cartilage
c. Cricoid cartilage and first tracheal ring
d. Epiglottis and hyoid horns
e. Thyroid isthmus and sternum
13. A patient undergoes surgical removal of the thyroid gland (thyroidectomy) within the muscular triangle. Postoperatively, the patient presents with a distinct change in vocal quality, characterized by hoarseness, due to accidental trauma to the:
a. Superior laryngeal nerve
b. Internal laryngeal nerve
c. Recurrent laryngeal nerve
d. Phrenic nerve
e. Ansa cervicalis
14. If a surgeon accidentally damages the external laryngeal nerve while ligating the superior thyroid artery at the upper pole of the thyroid gland, the patient will lose the function of which single muscle, resulting in an inability to create high-pitched vocal sounds?
a. Vocalis muscle
b. Lateral cricoarytenoid
c. Thyroarytenoid
d. Cricothyroid muscle
e. Posterior cricoarytenoid
15. A patient undergoes an extensive radical neck dissection for oral cancer. Postoperatively, when asked to protrude their tongue, the tongue deviates significantly to the right side, indicating iatrogenic damage inside the submandibular triangle to the:
a. Lingual nerve
b. Right hypoglossal nerve (CN XII)
c. Glossopharyngeal nerve (CN IX)
d. Mylohyoid nerve
e. Vagus nerve trunk
16. During the same radical neck dissection, a surgeon accidentally cuts the lingual nerve as it runs through the submandibular triangle space. This results in a complete loss of general somatic sensation affecting the:
a. Posterior one-third of the tongue
b. Anterior two-thirds of the tongue on that side
c. Hard palate mucosa
d. Soft palate and uvula
e. Vestibular gingiva of maxillary molars
17. If the chorda tympani nerve fibers traveling inside the lingual nerve are transected during a deep surgical approach to the floor of the mouth, the patient will experience a complete loss of:
a. Taste sensation on the anterior two-thirds of the tongue
b. General touch sensation on the chin skin
c. Motor control of the mylohyoid muscle

d. Secretomotor control of the parotid gland
e. Salivation from the labial minor glands

18. A patient experiences a severe tracking infection following a dental abscess in a lower molar. The pus perforates the inner table of the mandible *inferior* to the mylohyoid line, causing a massive, painful swelling that fills the:

a. Sublingual space
b. Submandibular space
c. Parotid space
d. Retropharyngeal space
e. Pterygomandibular space

19. If the dental infection fields breach the mandibular bone *superior* to the mylohyoid line, the subsequent abscess and fluid accumulation will isolate within the:

a. Sublingual space
b. Submandibular compartment
c. Retro-ocular matrix
d. Muscular triangle ceiling
e. Carotid sheath core

20. A 6-year-old child presents with a high fever, stiff neck, and difficulty swallowing. Radiographic imaging reveals a massive accumulation of pus in the potential space located between the buccopharyngeal fascia and the prevertebral fascia, formally diagnosed as a:

a. Sublingual abscess
b. Retropharyngeal abscess
c. Ludwig's angina space
d. Lateral cervical cyst
e. Carotid sheath phlegmon

21. A retropharyngeal abscess is a critical emergency because the infection can easily track retrogradely downward via gravity through the superior mediastinum directly into the posterior mediastinum, causing life-threatening:

a. Pleural effusions
b. Mediastinitis
c. Pericardial tamponade
d. Pulmonary embolisms
e. Esophageal ruptures

22. Ludwig’s angina is a severe, rapidly spreading bilateral cellulitis that involves the submandibular, sublingual, and submental spaces. The primary clinical hazard of Ludwig’s angina is rapid asphyxiation due to:

a. Selective clotting of the carotid artery
b. Massive edema shifting the tongue upward and backward, completely blocking the airway
c. Intracranial thrombosis of the cavernous sinus
d. Rupture of the internal jugular vein
e. Flaccid paralysis of the diaphragm muscle

23. Torticollis (wry neck) is a clinical condition characterized by a sustained, painful twisting of the neck that pulls the head toward one side. It is primarily caused by a pathological spasm, contracture, or fibrotic scarring affecting which muscle?

a. Trapezius
b. Splenius capitis
c. Sternocleidomastoid muscle

d. Omohyoid inferior belly
e. Platysma sheet

24. Congenital torticollis can occur in newborns due to a localized soft tissue injury or hematoma forming within the sternocleidomastoid muscle during a difficult breech delivery. If left untreated, this condition leads to permanent shortening of the muscle and secondary:

a. Facial asymmetry and restricted neck range of motion
b. Loss of vision in the ipsilateral eye
c. Clefting of the secondary palate
d. Accelerated closure of the anterior fontanelle
e. Odontogenic maxillary sinusitis

25. A patient presents with a deep laceration in the lower part of the posterior triangle of the neck. The surgeon notes severe, unresolving air bubbling inside the wound due to a tear of the external jugular vein. This occurs because the wall of the vein is tightly fused to the investing fascia, which prevents the vessel from:

a. Expressing clotting factors
b. Collapsing, creating a negative pressure hazard that can cause a fatal air embolism
c. Communicating with the internal jugular vein
d. Releasing platelets normally
e. Contracting its inner elastic layer

26. Winding downward across the superficial face of the scalenus anterior muscle inside the root of the neck is the phrenic nerve. Iatrogenic compression of this nerve during surgery results in temporary paralysis of the:

a. Intercostal muscles
b. Trapezius muscle
c. Diaphragm muscle on the affected side
d. Sternocleidomastoid muscle
e. Infrahyoid strap muscles

27. Scalenus anterior syndrome (or thoracic outlet syndrome) occurs when the structures emerging from the interscalene triangle are mechanically compressed. This compression characteristically involves the:

a. Common carotid artery and vagus nerve
b. Subclavian artery and lower trunks of the brachial plexus
c. External jugular vein and phrenic nerve
d. Internal jugular vein and ansa cervicalis roots
e. Sympathetic trunk chain links

28. A surgeon is operating inside the carotid triangle to clear an atheromatous plaque from the internal carotid artery (carotid endarterectomy). If they pull their retractor too tightly against the loop of nerve fibers crossing the superficial surface of the carotid sheath, they will cause a loss of motor function in the:

a. Muscles of mastication
b. Muscles of facial expression
c. Infrahyoid strap muscles (supplied by the ansa cervicalis)
c. Intrinsic muscles of the tongue
d. Intrinsic muscles of the larynx

29. If the superior cervical ganglion of the sympathetic trunk is compressed or damaged during a deep surgical approach to the prevertebral region of the neck, the patient will present clinically with:

a. Bell's palsy
b. Horner's syndrome
c. Frey's syndrome
d. Trigeminal neuralgia
e. Wallenberg syndrome

30. The jugulodigastric lymph node is a large, clinically prominent deep cervical node located at the intersection of the posterior belly of the digastric muscle and the internal jugular vein. Enlargement and tenderness of this specific node primarily indicates an infection or malignancy originating within the:

a. Apex of the chin
b. Lower incisor teeth
c. Palatine tonsil or posterior tongue
d. Skin over the occipital bone
e. Sublingual salivary gland

Answer Key Table (Contents & Clinicals - Set 2)

Q 1-8	Q 9-16	Q 17-24	Q 25-30

d	**b**	**a**	**b**
c	**a**	**b**	**c**
b	**b**	**a**	**b**
b	**b**	**b**	**c**
b	**b**	**c**	**b**
a	**b**	**a**	**c**
c	**a**	**a**	
d	**d**	**a**	